

# Ashish Gaurav

Senior Data Engineer | Data Integration | ETL/ELT | Databricks | Snowflake | Airflow

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## PROFILE SUMMARY

Senior Data Engineer with 4+ years of experience building healthcare ETL/ELT and enterprise data integration solutions using Databricks, PySpark, SQL, Snowflake, and Apache Airflow on Azure. Delivered batch pipelines processing **~1M records/day**, supporting **1000+ tables** and **200+ jobs/DAGs** within **3-hour SLAs**. Strong in source-to-target mapping, data modeling, DQ automation (**1200+ checks**), reconciliation, performance optimization, and secure PHI/PII data handling and sharing. Also built AI-enabled internal platforms for patient analytics and data modeling automation using Azure OpenAI.

## EXPERIENCE

**Optum** Mar 2025 – Present  
*Senior Data Engineer* Delhi NCR, India

- Contribute to architecture and design discussions for scalable healthcare data integration on **Databricks, Snowflake, and Airflow**, improving maintainability and operational stability.
- Optimize ETL workloads through Spark tuning, file layout/partition strategies, and Snowflake query optimization to meet **3-hour SLA** windows and reduce compute cost.
- Support orchestration and reliability for **200+ Airflow jobs/DAGs**, including dependency management, validation stages, retries, and failure-handling patterns.
- Implement CI/CD workflows using **GitHub Actions** to automate validation and deployment of Databricks pipeline/orchestration changes across environments.
- Mentor and onboard engineers through workflow walkthroughs, debugging support, and documentation to accelerate delivery and reduce production issues.

**Optum** Jan 2023 – Mar 2025  
*Data Engineering Analyst* Delhi NCR, India

- Built and maintained file-based ingestion and integration pipelines from **Azure Storage containers (TXT/CSV/Excel/Parquet)** into **Bronze (raw) tables**, followed by transformation into standardized integration/curated layers in Databricks and Snowflake.
- Delivered batch processing at **~1M records/day** while supporting **1000+ tables** and meeting **3-hour SLAs** through robust orchestration, validation, and operational monitoring.
- Designed unified target tables for multi-payer healthcare datasets by translating business requirements into scalable **SQL/Spark** transformations and reusable integration patterns.
- Delivered validated and curated datasets to **6 downstream enterprise systems** (including Curo, CDOS, and Cozeva), supporting both operational and analytical use cases.
- Developed an enterprise data quality framework with **1200+ automated SQL checks** (completeness, uniqueness, referential integrity, threshold validation, and reconciliation) to improve data trust and reduce defects.
- Generated outbound extracts (**TXT**) and delivered them to Azure containers for partner/system consumption with formatting standards, run-status tracking, and reconciliation controls.
- Supported secure PHI/PII data exchange using **ECG** for controlled sharing and compliance-aligned handling of sensitive healthcare data.

**Optum** Aug 2022 – Jan 2023  
*Technology Development Program - TDP* Delhi NCR, India

- Performed exploratory profiling of source datasets to understand schema, relationships, data patterns, and transformation requirements for ETL implementation.
- Supported development and testing of SQL-based transformations and source-to-target mappings; implemented validation checks for row counts and data consistency across layers.
- Assisted with debugging ETL execution issues during development and production releases by analyzing Airflow task logs and Databricks runs.

**Tredence** Jun 2022 – Jul 2022  
*Analyst* Delhi NCR, India

- Performed data analysis using Databricks and SQL/MySQL and developed Power BI and Excel dashboards to support reporting and insights.

## PROJECTS

**Member Insight – AI-Powered Healthcare Data & Patient Analytics Platform**  
*Internal Product / Analytics Platform*

Python, Flask, Snowflake, Databricks, Azure, Azure OpenAI, SQL  
2024 – 2025

- Developed an AI-enabled healthcare analytics platform enabling users to search member/patient data and run real-time analytics across **claims, pharmacy, eligibility, and clinical** domains.
- Implemented natural-language search and conversational follow-ups using **Azure OpenAI** with NL-to-SQL generation/execution on Snowflake and **grounded responses** based on actual query results, including error handling and validation.
- Built patient dashboards with multi-source data views, drill-down analysis, session management, and data-quality handling, supporting **1000+ internal users**.

**ModelForge – AI Data Modeling & Governance Assistant**  
*Data Architecture / Governance Enablement*

Python, FastAPI, Azure OpenAI, RAG, SQLite, Azure AD, JavaScript, HTML/CSS  
2024

- Built an AI-powered data modeling and governance tool that converts business requirements into reusable artifacts including **logical data models, DDL scripts**, data contracts, and data dictionaries.
- Implemented **RAG-based metadata reuse detection** using similarity search over existing models/definitions to improve standardization and reduce duplicated modeling effort.
- Added healthcare semantic mapping support (**FHIR/OMOP**) and governance workflows – approvals, audit trail, and compliance validation – with **Azure AD**-based access control.

**ETL Pipeline Monitoring & Operational Visibility**  
*Reliability / Operations*

Airflow, Databricks, SQL  
2024

- Improved operational visibility and triage across **200+ Airflow jobs/DAGs** via standardized run-status reporting and SLA tracking for batch pipelines.
- Analyzed recurring failure patterns and implemented monitoring-driven fixes, reducing MTTR and improving release stability for production workflows.

**TECHNICAL SKILLS**

- Programming & Querying:** Python, PySpark, SQL, Spark SQL, PL/SQL, MySQL
- Data Engineering:** Databricks, Snowflake, Apache Spark, Delta Lake, Apache Airflow, ETL/ELT, Batch Processing, Data Integration, Data Modeling, Data Warehousing
- Pipeline Engineering:** Airflow DAG Development, Workflow Orchestration, Source-to-Target Mapping, Transformation Design, Data Validation, Reconciliation, Root Cause Analysis
- Performance & Data Quality:** Spark Optimization, SQL Query Optimization, Partitioning Strategies, Data Quality Engineering, Automated Validation Frameworks
- APIs & GenAI:** Flask, FastAPI, Azure OpenAI, RAG
- Cloud & DevOps:** Microsoft Azure, Azure Storage (Containers/Blob/ADLS), Azure AD, GitHub, GitHub Actions, CI/CD Pipelines

**EDUCATION**

**National Institute of Technology Jamshedpur**  
*B.Tech Hons. – Electrical & Electronics Engineering*

2018 – 2022  
CGPA: 8.3/10

**CERTIFICATIONS**

- Databricks Certified Data Engineer Associate** – 2023
- Optum Specialized AI Dojo Certification**
- Data Science & Data Analysis with Python** – IBM